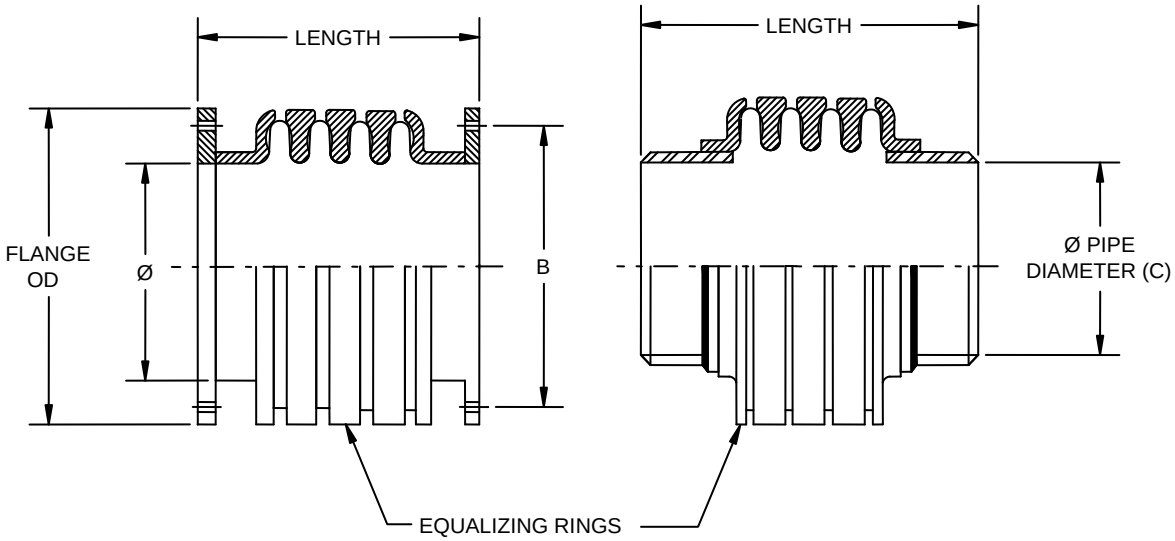
DETAIL SPECIFICATION SHEET

EXPANSION JOINTS, PIPE, METALLIC, SINGLE BELLOWS, CORRUGATED, REINFORCED

This specification is approved for use by all Departments
and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-17813.



DN	C-NPS inch (mm)	Flange OD inch (mm) 1/	B (bolt circle diameter) inch (mm) 1/
50	2 (50.8)	6 (152.4)	4.75 (120.7)
65	2.5 (76.2)	7 (177.8)	5.5 (139.7)
80	3 (76.2)	7.5 (190.5)	6 (152.4)
90	3.5 (101.6)	8.5 (215.9)	7 (177.8)
100	4 (101.6)	9 (228.6)	7.5 (190.5)
125	5 (127)	10 (254.0)	8.5 (215.9)
150	6 (152.4)	11 (279.4)	9.5 (241.3)
200	8 (203.2)	13.5 (342.9)	11.75 (298.5)
250	10 (254)	16 (406.4)	14.25 (362.0)
300	12 (304.8)	19 (482.6)	17 (431.8)
350	14 (355.6)	21 (533.4)	18.75 (476.3)
400	16 (406.4)	23.5 (596.9)	21.25 (539.8)
450	18 (457.2)	25 (635)	22.75 (577.9)
500	20 (508)	27.5 (698.5)	25 (635)
550	22 (558.8)	-	-
600	24 (609.6)	32 (812.8)	29.5 (749.3)

1/ See ASME B16.5, flange dimensions,150 pound.

FIGURE 1. Expansion joint, single bellows, reinforced.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:36Z
Check the source to verify that this is the current version before use.

NOTES:

1. Dimensions are in inches. Metric equivalents are given for information only.
2. The illustration is for identification purposes only and is not intended to restrict designs and shapes of fastening devices, which conform to the requirements and are within the envelope dimensions.

FIGURE 1. Expansion joint, single bellows, reinforced – Continued.

REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1.

Construction: Corrugated bellows, reinforced expansion joints shall be formed by one of the methods described in the EJMA Standards. They shall be provided with reinforcing or equalizing rings. Reinforcing rings, which support the root of the corrugation under internal pressure, are solid or tubular in cross section. Equalizing rings, which support the root and sidewalls of the corrugation under internal pressure are usually “T” shaped in cross section. Equalizing rings, sometimes called control rings, have diameters larger than the outside diameter of the corrugation and limit the maximum compression of each corrugation.

The classification of the expansion joints covered under this specification sheet are as follows:

Type II - shall be provided with reinforcing rings or equalizing rings.

Class 1 - shall have a single bellows.

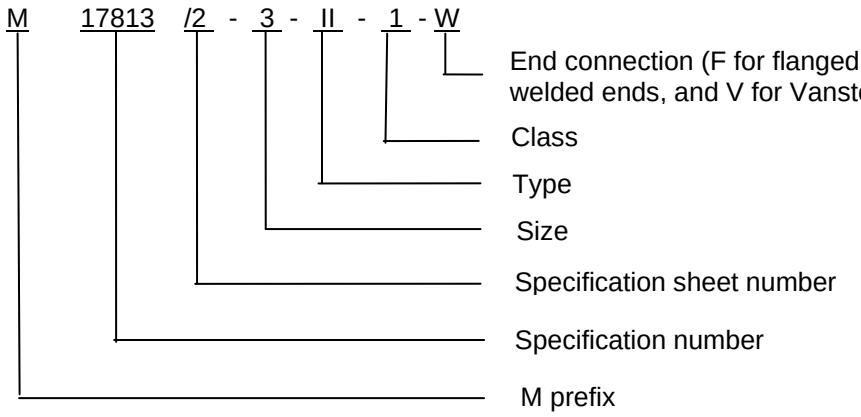
Type II joints designed for the absorption of axial movement shall, when required, be provided with suitable tie-rods, guide sleeves, or covers to prevent distortion of the expansion joint by weight of the reinforcement hardware.

Materials and finishes.

Materials and finishes for the bellows and flanges shall be in accordance with MIL-DTL-17813.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:36Z
Check the source to verify that this is the current version before use.

The Part or Identifying Number (PIN) consists of the letters M, the basic specification number, specification sheet number, size, type, class, pressure rating, and end connection.



PIN example: M17813/2-3-II-1-W

Referenced documents. In addition to MIL-DTL-17813, this document references the following:

ASME B16.5
EJMA Standards

CONCLUDING MATERIAL

Custodians:
Army - AT
Navy - SH
Air Force - 99
DLA - CC

Preparing activity:
DLA - CC

(Project 4730-2009-100)

Review activities:
Army - CE
Navy – MC, SA
Air Force - 71

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T10:36Z
Check the source to verify that this is the current version before use.